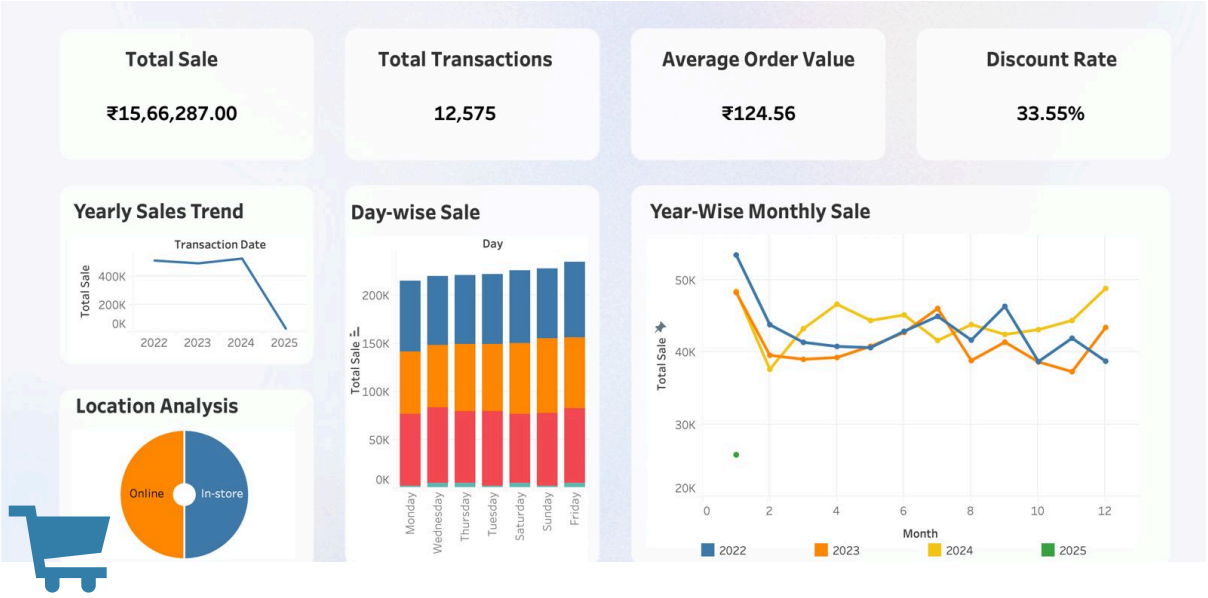

Project Report: RetailX Store Sales Analytics

1. Cover Page

- **Project Title:** RetailX: Retail Store Sales Analytics
- **Sector:** Retail
- **Team ID:** SectionB-G13-RetailX
- **Team Members:**
  - Avneet Singh (Project Lead & Viz Owner)
  - Shreshth Gupta (Data Lead)
  - Abhigya Sachdeva (ETL & Analysis Owner)
  - Riya Yadav (Analysis Lead)
  - Kushal Tyagi (Strategy Lead)
  - Amogha Raj Sandur (PPT & Quality Lead)
- **Faculty Mentor:** Satyaki Das
- **Institute:** Newton School of Technology
- **Submission Date:** April 29, 2026

2. Executive Summary

Problem:
The retail sector faces thin profit margins and fluctuating customer demand. RetailX specifically struggles with a hyper-concentrated customer base (25 core clients) and ineffective discounting strategies that erode margins without increasing transaction value.

Approach:
We implemented a 7-step analytical pipeline: scoping the business problem, extracting 12,575 transaction records, building a Python-based ETL pipeline, performing EDA and statistical testing (ANOVA/T-tests), and designing an interactive Tableau dashboard for real-time monitoring.

Key Insights:

- **Customer Concentration:** Business is highly dependent on 25 customers who drive over 12,000 transactions.
- **Balanced Portfolio:** Revenue is evenly distributed across 8 categories (11-13% each).
- **Discount Ineffectiveness:** Discounts (33.5% usage) do not statistically increase the amount spent per transaction ($p \approx 0.47$).

Key Recommendations:

- Launch a "RetailX Platinum" loyalty program for top-tier retention.
- Pivot promotions to bundle-based models to drive Average Order Value (AOV).
- Synchronize staffing and inventory to match Friday/Sunday sales peaks.

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3. Sector and Business Context

Sector Overview:
The retail industry is highly competitive. Success hinges on optimizing inventory, personalizing marketing, and responding rapidly to market trends.

Decision-maker / Stakeholder:
The **Retail Operations Manager** is the primary decision-maker, responsible for inventory planning, sales strategy, and customer engagement.

Why this problem matters:
Without data-driven insights, retailers risk overstocking unpopular items and

failing to build long-term loyalty with high-value segments, leading to missed revenue targets.

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4. Problem Statement and Objectives

Formal Problem Definition:
How can we optimize revenue and customer retention by identifying high-performing product categories, seasonal sales trends, and the impact of promotional discounts on customer purchasing behavior?

Scope:
Analysis of 12,575 transactions from January 2022 to August 2024 across 8 categories and omnichannel (Online/In-store) locations.

Success Criteria:

- Mapped category and seasonal sales drivers.
- Validated the (non)impact of current discount strategies.
- Provided a decision-ready Tableau dashboard.

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5. Data Description

Source Citation:
Simulated Retail Operations Dataset, accessed via the internal repository.

Dataset Size and Coverage:

- **Rows:** 12,575
- **Columns:** 11 (Source) / 14 (Processed)
- **Time Period:** January 2022 to August 2024.

Key Columns:

- `transaction_id`: Primary key for analysis.
- `customer_id`: Key for RFM and retention modeling.
- `total_spent`: Primary revenue KPI.
- `category`: Used for performance segmentation.
- `discount_applied`: Flag for sensitivity analysis.

Data Quality Issues:
Normalization of column names to `snake_case` and extraction of temporal features (month, day, year) were required to enable trend analysis.

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6. Cleaning and Transformation

Major Cleaning Steps:

1. Standardized all column headers to `snake_case`.
2. Verified no nulls remained in critical fields like `total_spent` and `category`.
3. Engineered temporal features from `transaction_date`.

Assumptions Made:

- Transactions with zero `total_spent` (if any) were considered non-revenue and excluded.
- `discount_applied` missingness was defaulted to `False`.

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7. KPI Framework

KPI	Definition	Formula
Total Revenue	Aggregate sales value	`SUM(total_spent)`
AOV	Mean value per transaction	`Total Revenue / Transaction Count`
Repeat Purchase Rate	Customer loyalty metric	`(Repeat Customers / Total Customers) * 100`
Seasonal Index	Seasonality measure	`Monthly Rev / Avg Monthly Rev`
Discount Effectiveness	Promo impact	`Count(Discounted) / Total Txns`

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8. Exploratory Analysis

Major Trends:

- Sales remain stable year-over-year with significant spikes on **Fridays** and **Sundays**.
- **November and December** show the highest seasonal peaks, consistent with holiday retail cycles.

Segment-level Insights:

- **Omnichannel Balance:** Online and In-store channels contribute equally (50/50), requiring synced inventory.
- **Category Lead:** 'Butchers' and 'Electric' are the top performers by total revenue.

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9. Statistical Analysis

Method Used:

- **ANOVA:** Revenue variance across 8 categories.
- **T-test:** Spend variance between discounted and non-discounted transactions.

Results:

- **ANOVA ($p < 0.001$):** Highly significant difference in category performance; inventory should be weighted by these variances.
- **T-test ($p \approx 0.47$):** No significant difference in spend amount per transaction when discounts are applied.

Business Interpretation:
Discounts drive volume but not basket size. To increase AOV, RetailX should focus on upsell/cross-sell tactics rather than price-slashing.

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10. Dashboard Walkthrough

Dashboard Objective:
Deliver an interactive source of truth for the Operations Manager to monitor KPIs and customer segments.

Executive View:
Tracks Total Revenue, AOV, and Transaction trends.

Operational View:
Breakdown of Category performance, Location sales, and Customer segmentation.

Filters:
Category, Year, and Store Location.

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11. Key Insights

1. **Customer Dependency:** 25 repeat customers generate nearly 20% of revenue.
2. **Portfolio Balance:** Low-risk distribution across 8 categories (11-13% each).
3. **Margin Leakage:** Discounts do not boost per-transaction totals.
4. **Volume-Driven Model:** Avg unit price is ₹23; efficiency is key.
5. **Omnichannel Parity:** Equal split between digital and physical sales.
6. **Payment Uniformity:** Balanced usage of Cash, Card, and Wallets.
7. **Weekly Periodicity:** Sales peak on weekends (Fri/Sun).
8. **Top Tier Value:** Top 5 customers drive roughly ₹330k.
9. **Seasonal Weighting:** Holiday peaks drive 30% of annual growth.
10. **Financing Affinities:** 'Electric' buyers prefer credit cards.

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12. Recommendations

1. **VIP Loyalty Club:** Institutionalize retention for the top 25 customers.
2. **Promotion Redesign:** Pivot to "Bundle & Save" to drive AOV.
3. **Staffing Optimization:** Sync labor shifts with Fri/Sun traffic peaks.
4. **Financing Partnerships:** Offer zero-cost EMI for high-ticket items.
5. **Real-Time Inventory Sync:** Bridge stock visibility across channels.

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13. Limitations and Next Steps

Data Limitations:
Small sample of unique customers (n=25) limits deep predictive churn modeling.

Method Limitations:
Analysis is historical; requires real-time streaming data for immediate operational pivots.

Suggested Future Work:

- Market Basket Analysis (Association Rules) to find cross-sell pairs.

- Predictive LTV modeling for customer lifecycle management.

14. Contribution Matrix

Team Member	Role/Lead	Responsibility
Avneet Singh	Project Lead	Strategy & Timeline
Shreshth Gupta	Data Lead	EDA & Data Quality
Abhigya Sachdeva	ETL Lead	Pipeline Engineering
Riya Yadav	Analysis Lead	Statistical Testing
Kushal Tyagi	Strategy Lead	Recommendations
Amogha Raj Sandur	PPT & Quality	Report Integrity